# Exit Ticket · reflect and retrieve

## Exit Ticket

A language such as C# can support both procedural and object-oriented programming.

## Use key vocabulary from memory

1. Explain paradigm in your own words.

2. Explain procedural in your own words.

3. Explain object-oriented in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.



RHS COMPUTING / OCR H446 / LESSON 229 DRAFT · TEACHER REVIEW

# Do Now · retrieval

## Programming paradigms

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

What is an object?

### Q2

What does a constructor do?

### Q3

What does a return statement do?

### Q4

What property distinguishes a pure function from a method that changes a shared counter?

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Procedural — Instructions and procedures operate on arguments or shared data. A function can be reused independently.

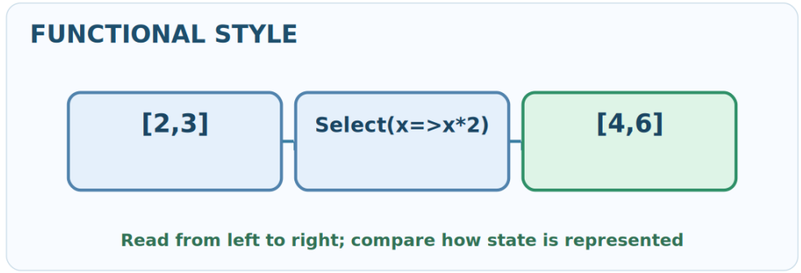
## Core idea 2

Object-oriented — The Account object combines state (Balance) and behaviour (Deposit). With inheritance, a subclass can reuse and specialise a class. A virtual method overridden in a derived class supports polymorphism: the actual object decides which implementation runs.

## Core idea 3

Functional style — Values are transformed using expressions/functions, often emphasising avoiding mutation of existing input data.

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.





# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Explain why `values.Select(x=>x*2)` can be considered functional-style programming, and one limitation of labelling C# purely functional.

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Classify a program using standalone functions with arguments and returns. Give one advantage.

## Question 2

Compare procedural and object-oriented account implementations, then explain how an overridden method on a derived SavingsAccount can be called through an Account reference.

Check that you have used the technical terms and explained your decisions.





# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

paradigm — Overall style for organising programs.

procedural — Organises a program around sequences of operations and subroutines.

object-oriented — Encapsulates state and behaviour into objects.

functional — Emphasises functions applied to values and reduced side effects.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

## Problem

Classify a function Square(x) that returns x*x without storing object state.

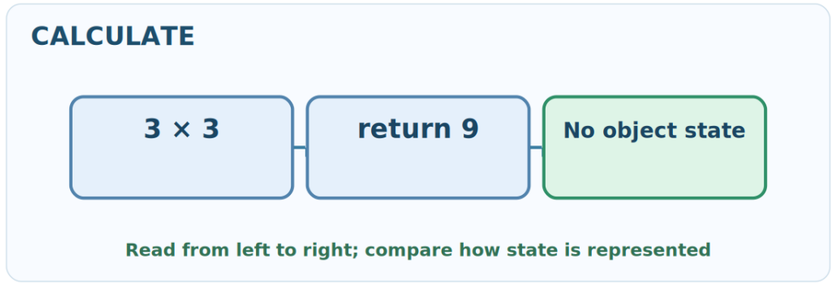
## Follow the worked method

1. Predict — It is procedural when used as a standalone function in a procedural programme.

2. Follow the declared and actual types — Both variables are declared as BaseMessage . Their actual objects are EmailMessage and TextMessage , each inheriting from the base class.

3. Output and design consequence — Outputs Email then Text . The caller can add another derived message class without writing separate type-checking branches.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Predict
The Score object holds Points alongside a method that changes its state.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.